

4 AI Formalism and Financial Competition

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4.1 Introduction

AI is marked by autonomously generated inventions and is self-assembling, which means automating duties that normally require human intelligence. AI is applied to a variety of business models. AI technology extends human competencies. Additionally, AI inventions produced by AAI systems will generate virtual results transferable into the human world as well via AI tools. Hence, technological innovation is modifying private markets around the globe, and new developments in digital technology have caused new opportunities for subtle and evasive forms of anticompetitive behavior by private corporations.¹

It is characteristic that the evolution of AI technologies has generated global connectivity, efficiency, and convenience; thus, in markets where there is fast technological transformation, even dominant companies are frightened by competitors' competencies, forcing them to repetitively buy and utilize tech devices and, on many occasions embracing anticompetitive practices that empower them to maintain their market advantage.

It has to be taken into account that countries have to become more competitive if they want to keep their economic position regarding other industrialized or developing countries and to react to encounters such as productivity growth, e-commerce, and technologies. Hence, competitiveness is grounded in the transformation of the economy into higher added value, productivity, and a more competent use of resources; thus, specialization is related to competitive advantages.² K. Khoo and A. Sng³ argue that "Competition Policy exists to maximise economic welfare, thereby creating an 'economically efficient' state of affairs," but economic welfare has to be achieved via particular competition rules applied equally

to all parties in a market because otherwise, economic welfare can encompass all methods of generating monopolies and oligopolies, resulting in harm to consumers in the end, as it is slowly happening with globalization. Also, Valdés⁴ assumes that

“free competition as a protected legal good seeks to harmonize the multitude of these freedoms of commercial competition existing in a civil society by way of limiting them to make them operational, and thus order them to the political common good.”

Corporations needing to remain competitive and progressive have to adjust due to global alterations by employing new management approaches, which means that rising international interaction, transformations in the technological space, rapid development of information technologies, and fluctuating consumer needs generate new challenges for corporations in a globalized world. Corporations have to align their personnel policies with broader strategic considerations, predominantly product market conditions and business strategy in a globally competitive environment.

It has to be taken into account that both antitrust and data privacy law affect corporations that keep and utilize personal data.⁵ Technology corporations have used privacy as a shield in order to explain allegedly anticompetitive conduct in antitrust enforcement cases. In *hiQ Labs, Inc. v. LinkedIn Corp.*⁶ HiQ challenged LinkedIn’s privacy policy, alleging that LinkedIn denied it access to publicly available LinkedIn members’ profiles, hindering hiQ from accumulating data to sell to its clients. In *Epic Games, Inc. v. Apple, Inc.*,⁷ a technology company used privacy as a shield against antitrust liability involving allegations that Apple illegally monopolized app distribution and in-app payments on the iOS App Store. In *Brunswick v. Pueblo Bowl-O-Mat, Inc.*,⁸ the Supreme Court stated that the purpose of antitrust laws is the protection of competition, not competitors. Justice Brandeis was a strong advocate of the democratic distribution of power and opportunity in the political economy, addressing how giant corporations’ actions affect socioeconomic problems such as low wages and

unemployment, pursuing to invalidate the established consumer welfare standard that is directed by price theory analysis.⁹

It is worth noting here that in platform ecosystems, interfirm conflict is restraining market power because the market controller can block the addition of new providers, which means that the adoption of emerging media technologies is postponed.¹⁰ Hence, a company, by having restricted access to technologies, cannot design a similar and possibly better alternative to the one offered by a platform.¹¹

It is obvious that there is a need to restore online competition. Also, competition between digital platforms increases the quality of privacy protection offered in the market. Moreover, it is argued that privacy is a factor of quality-based competition¹² but courts applying antitrust law cannot limit competition in the name of supporting other public policy interests such as public health, safety, or even privacy, in manners that are not specified by the legislation.¹³ Thus, antitrust/competition law promotes competition¹⁴ via formal legal competition rules identifying the particular economic elements grounding and advancing competition as such, not other public policy interests, even when those interests are undeniably lawful, which means that via competition all the other interests are promoted as well.

Innovations in ICT spark the transformation of society, consumer behavior, and workspace. Hence, digital transformation portrays the process of a corporation's prosperous adoption of digitalization. Moreover, the exposures of digitalized economies are tackled by corporations, and so the corporate digital responsibility (CDR) handles ethical problems initiated by digital transformation.¹⁵ Moreover, CDR is a new management notion to undertake responsibility in the digital economy voluntarily by shifting management and production systems along with accepting the governance mode.¹⁶ Digitization engrosses the automation of processes through the use of information and communication technologies; therefore, corporations incorporate technological changes since the inability to embrace technological advancements leads to competitive failures.

The modern digital economy is built on a platform of exploitation of regulatory gaps and liability shields. Platforms like Uber¹⁷ and Lyft and corporations like Amazon evade government regulation by manipulating

legal and regulatory structures to externalize costs and increase profits. Hence, as the digital economy grows, high-tech corporations are prioritizing wealth maximization above all else at the expense of workers, third parties, and society as a whole.¹⁸ Moreover, technology allows corporations to shift the risk and cost of excess labor or lost productivity to workers.¹⁹ It is worth noting that some of the connections between platforms and providers can be considered as that of a franchisor/franchisee structure.²⁰

There is a change from per se rules to a fact-sensitive methodology under the “rule of reason”. While in *Schwinn & Co*,²¹ the Supreme Court applied the per se standard, in *GTE Sylvania, Inc*,²² the Supreme Court applied the rule of reason and so, rejecting the per se standard.²³ Moreover, while in the 50s, the Chicago school caused no impact, in the 80s, the Chicago school has prevailed regarding its viewpoint that antitrust problems must be assessed via price theory.²⁴ Zekos²⁵ illustrated the advancement of formalism in competition via the presentation of the economics-based rule of reason, the more economics-and-effects-based method indicating that the “per se rule” applied first in price-fixing cases.²⁶ The Supreme Court²⁷ pointed out that “categories of analysis of anticompetitive effect are less fixed than terms like ‘per se,’ ‘quick look,’ and ‘rule of reason’ tend to make them appear.” Moreover, antitrust provisions against monopolization practices were not concerned with economic effects; rather, the statutory provisions contained in the Sherman Act were based upon the belief that “[...] great industrial consolidations are inherently undesirable, regardless of their economic results.”²⁸ Additionally, the antitrust laws were passed for “the protection of *competition*, not *competitors*.”²⁹ In fact, Zekos³⁰ has presented formalism denoting that the rule of law ideal is closely entangled with law’s traditional tendency toward formalism, and the rule of law ideal asserts that no one should be above the law; even the highest officials are subject to legal constraints, which means that judicial formalists prefer to check formal adherence to rules rather than the motives and effects of the actions.

It has to be taken into consideration that Formalism offers a basis for “objectifying” legal concepts³¹ and so, formalism generates strict definitional regimes in the form of cumulative necessary conditions for validity that, when satisfied and only when satisfied, meet the necessary

condition of negotiability which is compatible with that of defining competition law and not competition in economic terms. In other words, “Formalistic” approach encompasses a comprehension of setting the needed and sufficient conditions of operating law in many spheres such as competition.

It has to be taken into account that AI technology can be utilized to specify which elements are the elements of achieving competitiveness. Especially, the use of AAI technology will allow the continuous examination of the changing elements that, at the specific time, are factors of higher competitiveness. In other words, AAI technology by itself will investigate competitiveness, and in parallel it will automatically make the necessary advances in order to enhance competitiveness. Will AAI technology impose AI neo-formalism in competition?

4.2 Characteristics of Antitrust/Competition law

It has to be taken into account that there is no common competition law framework, and so, each country enforces its own competition laws, often with differing standards and procedures which means that a difficulty in enforcing antitrust policies has arisen, and so, inconsistencies in national regulations lead to unfair trade practices, monopolies, and market distortions generating anemic enforcement of competition laws and tolerating dominant corporations to engage in anticompetitive practices such as price-fixing and market monopolization.³² Also, it could be said that competition law is considered to be embedded in neoclassical economics. Furthermore, antitrust legislation is rooted in neoclassical economics, which revolves around the principles of price theory.³³

The antitrust laws were accepted against the economic concentration and the escalation of industrial trusts such as the railroad’s control.³⁴ To that extent, competition has to be a device for diffusing economic power and dissolving power in the market sphere, which means that complete competition must be protected. Therefore, competition law-based judgments should create a unified internal market governed by the rules of free and undistorted competition. Thus, antitrust should concentrate on

protecting and energizing competition, which is ultimately about economic outcomes achieved via a standardized competition process.

It has to be taken into account that antitrust laws aim to protect fair competition with the purpose of smoothing the playing field and lessening inequality. Moreover, fairness standards need clarity for daily business decisions, but competition forces corporations to tackle competencies, permitting them to advance the appeal of their products and services. It is deemed that unfair conduct is anticompetitive, and so, there is a need to standardize the elements denoting the context of fair, which means endorsing a formalistic approach that generates certainty about maintaining fair competition in the market. On the other hand, market competition is severe, upsetting corporations that fail to keep up with rivals, and so, economies of scale, along with the exploitation of network effects, give rise to large business corporations.

It has to be taken into account that the US antitrust laws do not levy restrictions on high prices or high profits, but they exclude transactions, agreements, and practices that harm or are likely to harm competition. Hence, antitrust law prohibits transactions, agreements, and practices that harm or are anticipated to damage competition. Moreover, antitrust law does not condemn the power to charge high prices when such competence is the outcome of greater product and business expertise. Therefore, the US Supreme Court has highlighted that, “as a general rule, businesses are free to choose the parties with whom they will deal, as well as the prices, terms, and conditions of that dealing,”³⁵ which means that the US antitrust laws have never prohibited high prices, regardless of their alleged unfairness.

Antitrust law and data privacy law are affecting the treatment of digital information.³⁶ Moreover, in the digital economy, data access can both drive competition and diminish privacy, due to the fact that breaching data privacy justifies anticompetitive conduct.³⁷ Also, it has to be taken into consideration that antitrust statutes speak plainly about economic perceptions such as monopolization, restraint of trade, and diminishing competition, thus embracing prohibitions on conduct. On the other hand, antitrust should focus on protecting and promoting competition, which ultimately concerns economic effects.³⁸ Judd Stone and Joshua D. Wright³⁹ say that “Antitrust’s central goal, though, is not the preservation of the

single-entity defense or Copperweld; instead, it is the maximization of consumer welfare through rules that account for error costs and administrative costs.”

It is argued that US antitrust law has been branded by economic functionalism, and European antitrust law has been distinguished by legal formalism. In other words, US law approves economically grounded analysis instead of formalistic thinking.⁴⁰ On the other hand, EU law is transitioning from a highly bureaucratic system with formalistic rules and procedures to a more effects-based system in which economics plays a fundamental role in shaping competition policy and enforcement.⁴¹ Furthermore, EU competition law has been characterized by legal formalism which means that the legality of certain business practices was determined on the basis of technical legal distinctions instead of their economic consequences.⁴² Also, the European antitrust community has assumed that the Guidance Paper reflects a movement away from a “form-based” analysis and toward an “effects-based” and functional economic analysis.⁴³ To that extent, Daniel A. Crane⁴⁴ argues that

“Europe employs legal formalism, and the United States uses economic formalism. Neither system employs an open-ended rule of reason type approach to loyalty rebates or is likely to do so any time soon. Both systems employ a mixture of functional and formal considerations.”

Up-to-date US antitrust jurisprudence is portrayed by elements of both open-ended rule of reason analysis, which induces detailed economic analysis, and more rule-bound approaches to particular problems of market power, where analysis is grounded on the application of one or two-factor liability determinants. Hence, the rule of per se illegality for price-fixing agreements is an instance of a liability determinant advancing a single-factor rule.

It has to be taken into account that the courts ascertain applicable normative rules and principles, both substantive and institutional, to guide antitrust decisions. Substantive rules are those broadening the fundamental objectives of antitrust law, which are incorporated in the “consumer welfare

standard.” Moreover, courts evaluate the facts in a case in light of the normative rules and principles and make a decision. It is argued that antitrust courts rely on various forms of a structured rule of reason in accomplishing both functions.

The per se rule is, in any event, intended to accomplish the same ends as the rule of reason in particular circumstances in which more extended factual inquiry is considered superfluous. Michael L. Katz and A. Douglas Melamed⁴⁵ argue that

“the antitrust rule of reason should be shaped, not just to guide correct application of existing law to the facts of the case (the second fundamental judicial function), but also to enable the courts to participate constructively in the common law-like evolution of antitrust law in the light of changes in economic learning and business and judicial experience (the first fundamental judicial function).”

In *Microsoft*⁴⁶ held “that the rule of reason, rather than per se analysis, should govern the legality of tying arrangements involving platform software products.”

It is argued that the courts read down the statutes to strike a balance between the aspirational and pragmatic impulses; thus, courts have considered the antitrust statutes as broad delegations allowing them to generate a pragmatic common law of competition.⁴⁷ Moreover, the Court has handled the Sherman Act as a common law statute.⁴⁸ As the Supreme Court explained in *Leegin*, “just as the common law adapts to modern understanding and greater experience, so too does the Sherman Act’s prohibition on ‘restraint[s] of trade’ evolve to meet the dynamics of present economic conditions,” which means that the new environment of trade brings forward factors that restrain trade and which did not exist in the prior conditions of trading in an economy, but the central essence of the rule in prohibiting restraints of trade is not altered.⁴⁹ Besides, Posner⁵⁰ specifies that currently the Sherman Act is not interpreted according to the framers’ understanding but judges have endorsed a more economic analysis.

To that extent, in a common law-like process, neither function can be well provided by picturing a static world in which normative legal standards and institutional considerations can be taken as fixed, and so both are treated as appropriate in the light of advances in economic learning and judicial experience that are important to the pending case, while at the same time giving proper deference to precedent.

It is worth noting that the principle/rationale of a rule is central, and merely the facts fulfill the content of the rule. Thus, the core conception of a rule does not change, but merely new events might fulfill its application as well. On the other hand, the generation of a new rationale for the same rule covering a further context means an alteration of the rule, which will encompass a new essence of the regulated matter.

Antitrust violations and, thus, anticompetitive conduct encompass actions that is not “competition on the merits,” diminishing the competitive constraints imposed by rivals. Furthermore, an increase in market power is triggered by anticompetitive conduct.⁵¹ It must be taken into account that that abusive conduct can go unnoticed as a result of a widespread “deference to innovation,” and numerous merger operations have been permitted for the same reason. Moreover, the need for an extensive interpretation of the concept of “substantial lessening of competition” is central to the rationale for authorizing concentrations.

The economic analysis shows that the assessment of the net consequences of specific conduct necessitates consideration of several market elements, embracing short-run and long-run outcomes that require complex rather than simple rules. Justice Brandeis⁵² refers to the rule of reason analysis, which determines whether a restraint suppresses and damages competition or simply regulates and promotes it, and encompasses an exhaustive factual review in which nearly anything could be appropriate evidence. Moreover, Brandeis⁵³ considered opportunistic intermediation as anticompetitive conduct opportunism, “self-interest seeking with guile,” denoting the exploitation of circumstances in pathways that are not automatically unlawful but are unfair.⁵⁴

Can antitrust law be utilized in achieving different social goals? It is obvious that antitrust law should not be exploited in pursuing numerous social objectives.⁵⁵ Hence, there is a need for standard rules achieving clear

tasks on various matters to ensure legal certainty and stability in trading. Rules must be used to achieve fair targets for society and not interpreted in a way that benefits an elite.

AI technology has led to an increase in the capacity of corporations to store, transmit, and process information. It is characteristic that information exchanges are not subject to per se condemnation; thus, the exchange of price data and other information among competitors does not always have anticompetitive consequences because, in particular settings, it augments economic efficiency, rendering markets more competitive. This means that such exchanges of information do not constitute a per se violation of the Sherman Act.⁵⁶ Also, corporations rely on live, readily accessible data and information from other corporations in making central business decisions; therefore, the exchange and use of information between horizontal competitors are linked to the competitive structure of the market in which these corporations compete. Consequently, the exchange of information between horizontal competitors can violate competition law provisions (Article 101, Section 1). Hence, the exchange of price-sensitive information may enable collusion between competing corporations⁵⁷ and information sharing can assist coordination.

It has to be taken into account that EU competition law has focused on economic welfare and, thus, the supposed “modernization” of EU competition policy denotes the promotion of the view that EU competition policy should concentrate on boosting economic efficiency and advancing consumer welfare, which means endorsing an efficiency-based, US-style conception. The “modernization” of EU competition law incorporates alterations in the interpretation and application of substantive competition law, endorsing the “more economic” approach.⁵⁸ Also, US antitrust policy remains principally concerned with the enhancement of economic efficiency and welfare.⁵⁹ Article 101 TFEU and Section 1 of the Sherman Act have been read to embrace two distinct centers of liability, and thus, the laws target both ex post collusion and the ex ante facilitation of such collusion.⁶⁰ It is worth noting that “economic collusion” denotes the situation where corporations in a given industry have prices that are higher than some competitive benchmark. It has to be taken into account that the modernization of the restriction of competition by object by the European

legislator and judges is a component of a general trend to enhance the detection and repression of anticompetitive behavior.⁶¹ Louis Vogel⁶² argues that “The return to formalism and the decline of the effects-based approach at the European level is basically the result of a misuse of the concept of restriction of competition by object.”

There has been a systematic and substantial move toward effects-based approaches in the DGCOMP decisions⁶³ by developing tools for the empirical measurement of the extent of economic analysis and legal standards in competition law enforcement, which means that legal certainty and predictability, which are the rationale of the law, are lost. It is characteristic that the choice of legal standards in antitrust enforcement is promoted to steer the assessment of possibly anticompetitive conduct, and cost of decision error principles are exploited in presenting a valuable and practical framework for scrutinizing the choice of legal standards for particular categories of conduct in competition law enforcement.⁶⁴ According to EU case law, the Commission grounds its description of the relevant market in cartel cases on the conduct of the participating undertakings, and so, the market analysis requires a coordinated conduct of two or more companies and an actual or prospective harm to other businesses or to consumers.⁶⁵ The new rules of the DMA result from the recognition of the inadequacy of traditional public antitrust enforcement in digital markets.

It has to be taken into account that the existing antitrust doctrines, rules, and antitrust enforcement “are too limited to protect competition adequately, making it needlessly difficult to stop anticompetitive conduct in digital markets” and growing market power.⁶⁶ It has to be taken into account that market power is the capacity of a corporation to profitably augment its prices above competitive levels and, as such, the exercise of market power lessens economic welfare, which means that an increase in market power echoes a drop in the effectiveness of actual or prospective competitors as constraints on a corporation’s conduct. Additionally, economic theories and empirical claims about the competition effects of certain practices are utilized for constructing antitrust rules by the courts. Moreover, Italianer⁶⁷ mentions that the US Supreme Court has plainly accepted that

“the categories of analysis cannot pigeonholed into terms like “per se” or...“rule of reason.” No categorical line can be drawn between them. Instead, what is required is a situational analysis moving along what the Court referred to as a “sliding scale.”

Furthermore, Douglas Melamed⁶⁸ argues that

“Maybe law reform efforts should look instead to core principles—antitrust is about prohibiting conduct that increases market power and cannot be justified by welfare-enhancing efficiencies—and to freeing the courts to decide cases by finding facts, instead of navigating a thicket of judges’ efforts to embody sometimes unsound, sometimes obsolete, and sometimes misunderstood economic propositions in legal rules.”

On the other hand, finding facts out of any legal rule cannot function because there is a need to endorse common elements in investigating the competition environment grounded in law in order to achieve legal certainty, permitting a stable economic environment. Courts apply legal rules in deciding cases and not economic principles, and factors that are too many according to economic theories and approaches lead to a divergence of elements; thus, legal rules are establishing fair competition in the market.

It turns out that modern competition law has converged too much on economic welfare and not enough on other values like fairness and equality, which initially stimulated antitrust legislation in the first place.⁶⁹ On the other hand, the law’s central role is to bring fairness and equality, which is not achieved in the case of competition law by exploiting economic interests and not by implementing certain formal laws that define their tasks formally. This allows for the achievement of the best outcomes concerning social, political, and economic matters due to their implementation in a fairly functioning market, rather than permitting capital to run the market according to economic principles outside any legal framework.⁷⁰

Also, Louis Vogel⁷¹ argues that “the law continues to allow the development of undertakings without subjecting them either to prohibitions which are unjustified in principle, or to the constraints of formalistic rules which excessively hinder their activity.” On the other hand, there is a need to define particular factors contained in standardized rules in order to achieve convergence on the matters causing anticompetitive contacts. In other words, strict formalism encompassing only basic interpretation of the formal content of a rule without any further analysis of formal rules connected to the regulated matter is not fulfilling the role of the law in implementing the rule of law and, consequently, competition law in regulating competition across all fields and topics in the market.

The utilization of the rule of law is well-kept when people understand that their lives will be wealthier if they follow the law. It must be taken into account that proof is the most indispensable element of the rule of law. Victor A. Bolden⁷² argues that “proof is the rule of law’s most essential element, a significance codified in legal rules, exemplified by legal theory, and reflected in the main source of belief in the rule of law, its effectiveness.” Due to the fact that evidence is the fundamental element illustrating the implementation of the rule of law in a jurisdiction, AAI employed in justice will provide the most accurate answer to a problem by examining all the possible data for the occasion.

It could be said that both EU and US competition rules are written in conceptually elastic language and are interpreted by employing a mixture of rules and standards. Stavros Makris⁷³ argues that

“Without openness, antitrust could become a formulaic and ineffective field of law unable to attain its core objective, the protection, and the promotion of competition. Yet, excessive openness can destabilize the Rule of Law or incite the instrumentalization of antitrust. In other words, integrity requires openness, but the latter, if excessive, can undermine the integrity of the law.”

On the other hand, by endorsing a kind of “sliding scale” mixture of formalism and effects-based analysis, legal uncertainty is generated which

brings forward a lack of competition enforcement. In other words, formalism allows the rightful interpretation of formal rules covering all the possible occasions of competitive conduct fitting in the wording of the formal rule but not arriving at an effects-based analysis over legal interpretation. To that extent, Geradin and Petit⁷⁴ mention that under a presumption of illegality, the assessment of such cases in the EU has relied on “old, formalistic legal appraisal standards, and (has shown) a reluctance to endorse a modern economic approach.”

Concerning matters of vertical restraints and monopolization and abuse of dominance, US antitrust law relies on extensive use of economic analysis and evidence in case-specific investigations; that is, it has relied much more on a rule of reason than on a *per se* approach. The major problem in using economic analysis concerning competition is grounded on the fact that economic analysis is not a single standardized process, but many different factors can be utilized in economic analysis, which means a total divergence in the factors acknowledged as influencing competition. Hence, an estimation of anticompetitive conduct grounded in standard legal rules brings convergence and certainty in assessing the competitive environment in a market. *Per se* rules minimize decision error costs, and so formalistic legal appraisal standards and presumptions of illegality bring legal certainty instead of any economic analysis grounded in unparalleled economic facts. Jan Broulík⁷⁵ argues that “*per se* and by object legal tests tend to be more rule-like whereas rule of reason and by effect tests are rather standard-like.” It could be said that on many occasions the application of the “rule of reason” is formalistic, since courts do not have to “balance” anything but simply rely on a clear-cut evidentiary process. Moreover, Crane⁷⁶ comments that “[r]ules, more than standards, provide *ex ante* notice of the law’s command and therefore enable the law’s subjects to conform their behavior to its dictates.” Also, Lemley and Leslie⁷⁷ agree that “*per se* rules...provide certainty.”

In short, general rules advance values that are important to any just law. Hence, competition rules must bring forward the values of competition and not any economic theory offering value for a particular party. To that extent, Justice Scalia has comprehended that “the trick is to carry general principles as far as it can go in substantial furtherance of the precise statutory or

constitutional prescription.”⁷⁸ Thus, per se rules on competition have to be employed in a formalistic way furthering the substance of the legal rule and not any economic rule to substitute a legal interpretation because we deal with competition law, and economic principles are applicable in identifying competition as an economic term. Moreover, standardized economic principles can be utilized as ex ante appraisal standards of the content of a formal rule, making the formalistic interpretation easier and, thus, generating predictability. Hence, in general, relying on economic theory to resolve issues of competitive harm is not counting on universally accepted dogma, but rather picking between competing economic theories.

In both occasions of legal and economic analysis, there is a reference to the content of the formal law in competition defining the terms, which means formalism is implicated in both cases. Justice Sandra Day O’Connor⁷⁹ noted that “[a]lthough the Sherman Act, by its terms, prohibits every agreement ‘in restraint of trade,’ this Court has long recognized that Congress intended to outlaw only unreasonable restraints” and so the Supreme Court first limited the range of the act by declaring that only contracts, combinations, or conspiracies in “unreasonable” restraint of trade violated it.⁸⁰ Of course, formalistically, courts based on the law interpret the intention of the legislature, elaborating on the degree of the prohibiting “restraint of trade.”

It has to be taken into account that an analysis of the competitive benefits and harms of a practice unavoidably presents some legal uncertainty as long as the analysis is not grounded on ex ante economic formality producing predictability and certainty.⁸¹ It is worth noting here that modern antitrust law is moving away from rules encompassing ex ante limited factor liability determinants and toward standards including ex post multi-factor liability determinants. Likewise, economic modeling generates unpredictability and per se rules “give clear guidance” to businesses as to the lawfulness of their conduct.⁸² Meanwhile, the EU Court of Justice mentioned in *Budapest Bank*⁸³ that business conduct hampers competition by object only when it is “so likely to have negative effects...that it may be considered redundant, for the purposes of applying Article 101(1) TFEU, to prove that it has actual effects on the market.”

Competition laws in the USA, along with Europe, rely on individual assessments of competitive effects in every case, and so, it cannot be anticipated by businesses to whom competition law is directed which means that unpredictability damages the objective of competition law by bringing about over along with under-deterrence.⁸⁴ In line, Jan Broulik⁸⁵ specifies that “those who invoked predictability of competition law as a reason to modernize competition law by making it more economics based rarely warn against unpredictability caused by too much economics in enforcement.” It seems that EU competition has moved from formalism toward an “economic approach which is based on the effects on the market”⁸⁶ and so by endorsing an “effects-based approach,” a real “verification of competitive harm” is needed.

It has to be taken into account that the lack of predictability in enforcement hampers the preventive function of competition law. It has to be taken into consideration that competition law enforcement has to be grounded on clear legal rules that pre-exist the factual conditions of the case, generating legal certainty.⁸⁷ Also, encompassing economically informed legal categories into ex ante legal rules strengthens the formalistic application of law by furthering the spectrum of covered occasions and, thus, upgrading predictability.

It is characteristic that the Supreme Court has engaged in overt antitrust anti-textualism, causing a concurrent drop in both antitrust enforcement and regulatory power.⁸⁸ M. A. Katz argues⁸⁹ that “the appropriate relationship between antitrust and regulation is neither adversarial nor cyclical, but symbiotic.” Hence, deregulation and diminished enforcement do not cause greater effectiveness and more robust competition; rather, there is an upsurge in the concentration of market power, allowing a few corporations to gain by building monopoly power across businesses.⁹⁰

It seems that de-formalization is the latest approach in US competition law. By contrast, presently it seems that courts endorse the anti-text pattern⁹¹ and so, consumer welfare is endorsed as the exclusive goal of antitrust.⁹² John M. Newman⁹³ argues that “moving beyond the narrow confines of output-based analysis enables the application of a more coherent, practical, and efficient antitrust framework.”

Has the shifting of the approach concerning competition law of the US Supreme Court from formalism toward consumer welfare generated lawful implementation of competition law or a monopolization of the market relying on politics? It could be said that AAI technology will choose an objective implementation of competition law toward competitive success grounded on legal certainty and predictability, leading to an open market for competition and preventing the emergence of monopolies and oligopolies. Does the goal of antitrust law maximize the output of markets? Of course, the role of competition law is to achieve the fair functioning of the market, which means open market for any potential candidates to operate on equal terms and not maximizing economic outputs regardless of wasting global resources or other economic techniques of gathering global wealth in the hands of an elite against the well-being of the globe. It is worth noting that the maximization of wealth and consumer welfare as goals of antitrust law has led to market power encompassing the power to control competition.⁹⁴ Filippo Lancieri⁹⁵ et al. argue that

“The decline of antitrust enforcement from the 1970s to the present was not achieved through legislative reform in response to public demand. It was the result of decisions made mostly in the shadow by politically unaccountable officials—judges and regulators—whose views of antitrust at the time of their appointment were (in most cases) not publicly known or perhaps even clear in their own minds.”

As discussed earlier, US courts developed the “per se” category involving “agreements whose nature and necessary effect are so plainly anticompetitive that no elaborate study of the industry is needed to establish their illegality” and the “rule of reason” category linked with “agreements whose competitive effect can only be evaluated by analyzing the facts peculiar to the business, the history of the restraint, and the reasons why it was imposed.” Like the per se rule, a presumption of illegality is also drawn with regard to object-type infringements, denoting that in both instances

formalism to an extent comes out.⁹⁶ Besides, Kenneth Khoo and Jerrold Soh⁹⁷ argue that

“the quasi–per se nature of the EU competition regime’s treatment of such information exchanges may lead to significant wrongful convictions, a major cause of economic inefficiency...the increase in error costs following from such formalism is likely to outweigh any countervailing decrease in administrative costs. Thus, for a jurisdiction considering an appropriate framework to establish generic competition laws, there are good reasons to adopt a U.S. style rule of reason approach toward all instances of information exchange... highlight an instance of how overly formalistic rules may be inefficient, subverting the welfarist objective of antitrust laws.”

It could be said that the rule of reason suffers from several setbacks such as vagueness, unpredictability, high costs of litigation, and problems in gathering facts; thus, the rule of reason presents too little guidance to the business community regarding where the boundaries of allowable conduct lie. Per se rules are presented as a means of tackling the disadvantages of rule of reason analysis because per se rules enhance the predictability of analysis and litigation outcomes, thereby progressing business certainty and administrative effectiveness. Moreover, the interests of administrative effectiveness in antitrust regulation are inducing.⁹⁸

It has to be taken into account that the rule of reason has been criticized for offering too little guidance to the business community concerning where the boundaries of permitted conduct lie⁹⁹ but, on the other hand, formalism should provide simple legal rules offering legal certainty.

It is worth noting that the per se rules allow businessmen to predict in any actual case what courts will decide to be legal and illegal under the Sherman Act.¹⁰⁰ Hence, litigation should not blur clear rules into vague standards¹⁰¹ and courts and litigants facing vague standards seek to convert those standards into clear rules. Thus, legislation and courts have to provide

definitions of price fixing unrelated to the underlying standard protecting the predictability and certainty of the per se rule.

It seems that antitrust law's per se rules simplify analysis. Sean P. Sullivan¹⁰² argues that

“efforts to convert vague antitrust standards into clear rules will rarely succeed without abandoning the underlying standards that the rules were meant to simplify. Second, I conjecture that failed efforts at simplifying antitrust will often have the opposite effect—increasing the apparent complexity and vagueness of this law. If these conjectures are correct, then the search for simpler rules could be not just unproductive but counterproductive in antitrust law.”

The fundamental goal of antitrust law is to protect the economy against harms associated with cartels and monopolies. Moreover, the law should spell out and prohibit a specific set of abusive business practices¹⁰³ which are to be regarded as anticompetitive, thereby identifying ex ante and in a formalistic way the constraints against competition. This means clearly expressing the content of the competitive process as the legal standard for antitrust law.

It is characteristic that product industrialists have employed proprietary software and predatory design to preclude independent servicers from repairing their products and, thus, obliging customers to repair via manufacturer-authorized servicers.¹⁰⁴ On the other hand, the Right to Repair Movement advances competition within repair aftermarkets by reducing costs by neutralizing manufacturers' efforts at monopolizing repairs aftermarkets.¹⁰⁵ It is worth noting that the Right to Repair Movement and antitrust law are promoting market competition.¹⁰⁶

Market concentration and vertical integration allow monopolization, while fair competition generates an AI marketplace with less biased applications winning consumer approval. T. N. Narechania and G. Sitaraman¹⁰⁷ argue that “Because antitrust enforcement operates ex post and on a case-by-case basis...antitrust doctrines have been narrowed over

the last forty years, rendering underenforcement more likely in a sector that seems structurally inclined towards concentration,” which means that anticompetitive behaviors endure and market power has become more intense. Hence, by endorsing antimonopoly rules functioning ex ante as market-shaping devices prevent harms by modeling market structure.

It could be said that the DMA advances barriers to entry for startups and, as a result, dampens investment in emerging digital ventures, which means that the DMA restrains motivation to take the risks needed for business achievement, eventually obstructing the advancement of revolutionary AI expansions. Hence, it seems that by advancing regulatory clarity and certainty, the EU will promote a more advantageous environment for innovation and competition in the EU digital market. To that extent, K. Afuwape¹⁰⁸ argues that

“The shift in the DMA from ex-post to ex-ante rules, using per se antitrust prohibitions, fails to differentiate between conduct that fosters competition and conduct that hinders it. Consequently, the DMA overlooks the benefits of economic efficiency and conflicts with the EU’s principle of proportionality.”

On the other hand, competition law has to follow the principles of law based on standardized content of rules and not grounded on changeable economic factors, generating uncertainty instead of legal certainty leading to economic advancement and not economic monopolies.

Competition laws intend to deliver free, fair, and undistorted competition, which is achieved via ex ante standardized rules rather than inserting economic evaluation ratios, as that of the vague “consumer welfare” filter. It seems that US antitrust law and, to some extent, EU competition law include a “consumer welfare standard” (“CWS”) filter.¹⁰⁹ It could be said that by adding or subtracting legal or empirical facts in interpreting legal rules, one assimilates uncertainty in the essence of competition law and market competition, which means that legal formalism is required. Competition law principles should be standardized and revised occasionally in order to incorporate new developments, but always in a

formalistic way and not be “a complex aggregate that incrementally consolidates dozens of epistemological choices”¹¹⁰ and so, avoiding uncertainty. It seems that the ECJ in the AGCM¹¹¹ decision incentivizes innovation in advancing market economy, regardless that the law is technology neutral, which means formalistic standardized legal principles expressing standardized economic rules and not vague economic theorizing in order to establish legal certainty upon which market competition has to be developed. Consumer interests, and behind the curtains those of monopolies, should not be the objective of competition law but only the establishment of fair competition grounded on formal legal competition rationales, thus ensuring real market competition for all participants, achieving quality economic outcomes for society and not for global monopolies.

4.3 Globalization and Competition

It is worth noting that globalization has transformed the basis for competition, and so, corporations compete in global markets which means the need for executing global strategies to sell their goods, acquire raw materials and intermediate products or services in order to control costs in global markets. Additionally, governments have to improve competition through reforms in antitrust prosecution instead of allowing market concentration with adverse effects on competition.

There is an increase in concentration and profitability driven by corporations’ capability to extract higher operating margins via higher markups. Also, the current market concentration via AI technology produces extraordinary profits and a reduction in investments in innovation. Additionally, high-tech corporations overshadow the stock market and make it difficult for startups to raise funds, thus prohibiting competition via innovation in order to achieve sustainable economic growth. Big AI corporations are acquiring smaller AI startups and projects in the early stages of their progress, thereby abolishing any future prospective competition, which means that these acquisitions reveal the desire of these corporations to pile up intellectual property, datasets, and highly trained human resources in the AI arena and so, competition is suffocated.¹¹²

It has to be taken into account that high-tech corporations have increased their market power, thereby diminishing market competition, which means not only higher price markups but also intensifying income inequality, generating a concern for governments due to geopolitical consequences. In line with this, D. Martínez and P Bunkanwanicha¹¹³ argue that “governments should put more effort into strengthening market competition to deal with these challenges rather than promoting political-driven national champions to compete with superstar firms.”

It seems that the revenues of high-tech corporations keep pace with governments, and so, a massive economic concentration leads to political power being in a few families’ hands which means that the interface of economic and political power is a threat to the well-functioning of a free market global economy. It is characteristic of corporations find tactics to exercise their political power by lobbying government officials to gain benefits such as more favorable regulations that permit them to freely implement their market power or receive government contracts.

It is characteristic that high-tech corporations are exploiting their monopolistic power over customers, suppliers, and employees to retain their extraordinary profits and, via their political power, prolong their dominance at a high cost to society. Hence, the concentration of power in a few corporations has caused a rise in income inequality, with a high return for shareholders at the cost of a deterioration in labor part.¹¹⁴ To that extent, D. Martínez and P Bunkanwanicha¹¹⁵ argue that

“Superstar firms have developed their competitive advantages levered in technological improvement and innovation, augmenting sales at a much higher rate in relation to workforce increase. This effect has generated wealth creation for superstars’ shareholders at the cost of the labor force, increasing inequality. This unequal structure in which superstar firms are far from the laggard firms will continue penalizing labor share and generating inequality. Regulators need to look for tools to improve competition within these

industries, such as facilitating access to funding for start-ups and reducing regulatory and artificial barriers to competition.”

Governments have to break down barriers artificially built by corporations with high economic and political power. Hence, there is a need to properly supervise market power via a plain disclosure of corporate activities. On the other hand, dominant corporations seek to uphold their dominant position via acquisitions, which is a legal way of advancing the status of a corporation. Thus, achieving high standards of growth via innovation and productivity in fair competition is the motive for business. Fabienne Fel¹¹⁶ illustrates that “the next generation of products manufactured in France will replace current foreign supplies, which will have become obsolete.” Hence, there is a need for France to improve competitiveness via its high prospective for innovation and modernization.

Can AAI technology prohibit monopolies? First of all, the current AI systems are utilized by humans in order to achieve monopolies more easily and contribute to preserving the downsides of the current human society regarding competition, labor, health, etc. It could be argued that AAI corporations will have large structures in order to be functional in controlling vast volumes of data to keep competition and other social aspects of life at an objective level, enabling the fair rolling of AAI society. Moreover, it has to be understood that AAI humanoids will have different needs and social lives than humans and so AAI systems might be able to fulfill their needs easily as long as they are merely electronic entities dependent on the central AAI systems for survival. On the other hand, if AAI society is developed along a parallel path to that of humans’ society and merely functions at a different technical level, then the solutions for AAI disputes will be more complicated and not easily controlled, which means that the malfunctions in human society will emerge as well.

Banks are endorsing platform-based business models that are more customer-oriented and offering user-friendly services. The “platformization” denotes a transfer from individual products/services to platforms as intermediaries for transactions and, in doing so, generating value design courses, which means advancing the formation of ecosystems

in delivering products and services within the platform by producing value for both its users and the platform owner.¹¹⁷

It has to be taken into account that AI innovation and digitalization have generated new tasks for incumbent banks by forming new market standards and altering their competitive landscape due to consumers' expectations and uses. In line, R. Coeurderoy and M. Guilhaon¹¹⁸ argue that "Banks are facing paradoxical pressures between conformity to their core activities in which they are perceived as legitimate, and differentiation that could help them renew their competitive advantage."

If the competitive process functions well across an economy, it drives overall investments, innovation, and productivity growth. Also, competition is valuable to customers, leading to long-term growth. In other words, competition should first focus on the employment of a standardized competition process followed by all corporations, and via fair competition process, the whole economy will advance properly by distributing all the gains to society and not allowing competition to cause monopolization of various sectors of production and services.

Taking into account that market power has augmented across the global economy, competition law has failed to ensure competition for the participants; thus, competition law enforcement has been abandoned. Additionally, it should be considered that dominant corporations are shaping the public debate by promoting competition economics rather than competition law enforced in a formalistic way. The uncertainty generated by competition economics should not be normatively acceptable in the assertion that competition economics actually benefit consumers which in great numbers is not true if quality matters are counted in the evaluation of consumers' benefit. In other words, competition law cannot be treated in terms of competition economics.¹¹⁹

4.4 AI and Competition

It is characteristic that the digital revolution comprises a great success of the market economy by stimulating vibrant competition in the markets.¹²⁰ The advancement of the digital markets has generated a disproportionate

concentration process, and so a few large platform undertakings have reached a dominant position in some markets becoming “gatekeepers.”

It has to be taken into account that sovereignty with reference to digital markets focuses on the control of the flow of data, the capability to set standards in achieving interoperability, along with the implications of the acquisition of significant economic power by digital corporations.

The achieved market power permits Big Tech to expand their activity in other sectors and build synergies between different products and services, triggering a sort of “ecosystems” where new entrants cannot enter, which means that it is easy for them to misuse their dominant position. On the one hand, antitrust laws protect fair competition, but owing to the lack of fairness standards and clarity, corporations exploit efficiencies that could allow them to expand the appeal of their products and services. On the other hand, Big Tech monopolies have abused their control over access to markets by choosing winners and losers throughout the economy in their favor by discriminating among third parties, entering new markets, establishing their monopolies, and misusing their power.¹²¹

It seems that states have emerged as monopolists because of privatization. Gregory Day¹²² argues that “collusions among private actors and states have not only oppressed marginalized communities, but also imposed society’s most durable monopolies. To uphold principles of federalism, but also rectify state oppression, courts or Congress must abrogate state action immunity in its entirety.”

New technology giants have won market dominance via the collection, analysis, and synthesis of data, indicating that the increasing dependence on digital technology through data gathering is a precondition for economic participation. Moreover, the US Department of Justice¹²³ recently specified that American Big Tech corporations pose “a threat to open markets and competition, with risks for consumers, businesses, innovation, resiliency, global competitiveness, and our democracy.” It must be taken into account that the digital platform business model has grown, and there is a need for the economic analysis of platform competition.¹²⁴

Digital platforms use AI to personalize their services or products, thereby augmenting their installed base. Moreover, platforms activate the data network effect, and by utilizing AI, data collected by certain

technology corporations gives them a competitive advantage that is nearly impossible for rivals to match.¹²⁵ To that extent, Big Tech corporations have become gatekeepers in digital markets, making their positions unquestionable due to the function of platforms generating operating systems, app stores, online marketplaces, and search engines. These markets have strong increasing returns, which are instigated, among others by network effects. Moreover, multiplying returns lead to competition being for the market rather than in the market¹²⁶ and so, in a killer acquisition, the acquiring undertaking tries “to discontinue the development of the targets’ innovation projects and pre-empt future competition.”¹²⁷

Taking into account that digital platforms set restrictions on participation, development, or use across their clear-cut roles, platforms clarify how “open” they are to users and developers. Thus, the utilization of AI technology can achieve the best point of balance in order to have the best possible outcome due to the fact that the highest volume of data on the openness can be taken into consideration by an AI system. Also, it seems that acquisitions lead to lower than optimal levels of openness. To that extent, J. Manuel Sanchez-Cartas¹²⁸ argues that

“a platform always has an incentive to acquire or imitate complementors. Which one is preferred depends on whether the increase in platform value (acquisition) offsets the market expansion effect (imitation)...acquisitions reduce openness and welfare but may generate more valuable complements while imitation increases openness and welfare but may harm third-party developers.”

As mentioned earlier, Big Tech monopolies have abused their control over access to markets by establishing their monopolies throughout the economy, utilizing and bending the playing field in their favor, and expanding into new markets by utilizing their power in many ways.¹²⁹ It is argued that Big Tech corporations, as a result of their market position, shape terms and protect and extend their market power.¹³⁰ Hence, it is characteristic that

digital companies have expanded the range of their economic activities through internal growth and acquisitions.

AI expands the situations in which known forms of anticompetitive conduct, mainly conscious parallelism or tacit collusion, take place. Moreover, the use of algorithms brings newer types of anticompetitive conduct, confronting traditional antitrust orthodoxy with new elements such as price discrimination, co-opetition, data extraction, and data capture. Additionally, deception is integral in algorithmic markets, prompting consumers to take part in manipulative transactions.

AI and digitization increase competition in many ways, but simultaneously, altering technology brings new sources of concentration, embracing powerful network effects. The digital economy boosts large digital platforms that rely profoundly on direct and indirect network effects to grow, holding high market shares and thus altering competition. It could be said that AAI technology, by implementing AAI competition law formalistically created, will automatically intervene, canceling any anticompetitive conduct of digital platforms and thus allowing an objective application of competition law to accomplish fair competition.

Current antitrust means tackle some prospective anticompetitive outcomes of information exchange linked to AI platforms if they act as a hub for systematic information exchange.¹³¹ A correct definition of the relevant market is a prerequisite for any assessment of the effect of a merger on competition.¹³² Price fixing and collusion could be detected in the communications between businesses affecting competition, but the task of discovering detrimental price behavior becomes more difficult with the use of AI algorithms for setting prices, leading to unwanted price behavior, sometimes even unintentionally. Additionally, the use of advanced algorithms to set prices and adapt product functionality increases opacity.

There is a need for formality in the rules applying to the utilization of AI systems and the consequences of any failure. In other words, there is a demand for formalism in regulating AI implementation according to the seriousness of the consequences of any failure.

4.5 Formalism and AAI Neo-Formalism in Competition

Formalism is stated as a group of opinions about legal interpretation that embraces both textualism and other opinions considered correct interpretation in line with the original legislative intentions.¹³³ It could be said that formalism denotes obedience to the rules, which means that formalism should provide legal rules offering legal certainty. K. Llewellyn¹³⁴ stresses that formalism is based on “canonically” written language of rules. Moreover, M. Horwitz¹³⁵ argues that formalism is a way of denying the diagnosis of influential functions of law. Furthermore, Tushnet,¹³⁶ tells us that formalism incorporates an artificially restrictive array of interpretive choices.

Also, R. Unger¹³⁷ specifies that formalism belongs to a compelled and relatively apolitical decision-making process. However, law, with its wording, identifies the regulated matter under particular terms, and politics cannot intervene in order to achieve its desired decision, promoting particular interests via a special economic theory. In addition, according to Kennedy¹³⁸ formalism is grounded in the viewpoint that rule application is mechanical, and so, the mechanical application is fair. Felix Cohen¹³⁹ finds that formalism is a “philosophical basis” for “objectifying” legal concepts. Likewise, Strauss¹⁴⁰ says that formalism is a denial of acknowledging the practical aftereffects of judicial decisions. It is argued that decision-makers¹⁴¹ take all factors they believe to be relevant into account, or at least that they feel obliged to reach “reasonable” results, whether or not the language of the rule points in that direction.

It is worth mentioning that formalism is a practical working instrument used in law, economics, and other disciplines.¹⁴² Hence, it has to be stressed that competition formalism amalgamates formalistic rationales from law and economics. Also, it is argued that formalism can be implemented in both law and economics.¹⁴³

“Formalistic” is a normative architecture that restricts the scope of “family likeness” open-ended definitions to sets of needed and appropriate conditions. Thus, formalism involves a method of construction by

disconnecting legal application from normative or policy considerations. Jonathan Yovel¹⁴⁴ argues that “formalist construction appears in degrees and can be functionally motivated.”

It is characteristic that views of formalism implicate the notion that rules constrict the choice of the decision-maker, and so “formalism” denotes the decision-making according to rule, which means that it is the very idea of decision-making by rule. Hence, there is a need for precise rules in order to achieve certainty via predictability. Otherwise, there will be a divergence on regulated matters such as in competition case law where various economic calculations are considered as the picture of anticompetitive conduct, leading to a relaxed applicability of competition rules. It is representative that formalism is unfolding an over-reliance on canons of decision¹⁴⁵ and so formalism reveals that legal decision-making fulfills instrumental functions.¹⁴⁶

It is argued that legislation on any subject will encompass gaps and errors that judges fill and correct when cases bring to light problems that the legislature could not and/or did not foresee.¹⁴⁷ Hence, the legislature may not predict every situation to which its statutes will apply, and so, writing in general language, it cannot cover every occasion that, with a rightful interpretation, will be enclosed and regulated by the content of the rule. Statutes are the instrumentalities by which policymaking institutions accomplish their mission by determining society’s organization, and courts are implementing statutes. There is a need for the typical *legal* way of interpreting doctrinal categories and structures of the law. On the other hand, it could be said that textualism offers certainty in the law, predictability, and respect for the rule of law. Thus, legislators enact and judges interpret texts without incorporating extraneous matters. To that extent, Justice Scalia explains that the court’s “regular method for interpreting the meaning of language in a statute” is to “first, find the ordinary meaning of the language in its textual context; and second, using established canons of construction, ask whether there is any clear indication that some permissible meaning other than the ordinary one applies.”¹⁴⁸ Moreover, Daniel A. Farber¹⁴⁹ argues that “the vices of formalism are excessive confidence in the power of ‘the word’ and excessive distrust of the ability of judges to exercise good judgment.” Judges cannot insert

meanings or economic or social effects that alter the central understanding of a term encompassed in a formal law because the implementation of the law has to achieve the goals attributed to the law by the legislature, and so, judges cannot establish via their judgments a new law but merely elaborate on the central rationale of the law without altering the task of the legislature.

It is worth noting that judicial power is employed for the rationale of giving effect to the will of the legislature, being the will of the law and not that of the judge. Moreover, textualism generates greater certainty in the law, which means greater predictability and respect for the rule of law.¹⁵⁰ Also, it could be said that any jurisdiction adopting laws cannot survive without the belief that words express apparent meanings and without the loyalty of legal arbiters to conform to those meanings.¹⁵¹ In line with this, Daniel A. Farber¹⁵² argues that “formalist interpretation, ultimately, relies on a faith in the raw power of the word to communicate, as if the perplexities of statutory interpretation were due merely to legal sophistry.”

It has to be taken into account that the antitrust laws apply to commercial conduct touching interstate commerce and so, they have to be appropriate for a vast and ever-changing array of conduct and conditions, the consequences of which might be apparent only after extensive, detailed, and particular factual inquiry. Hence, Congress has delegated to the courts the application of both the normative standards to be applied in evaluating conduct and the process by which courts decide whether these standards are violated, which means that this delegation “permits the law to adapt to new learning.” It could be said that none of those terms are defined in the statutes and so, enabling flexibility. On the one hand, the imprecision of the statutes is probably a good thing, but on the other hand, imprecision brings uncertainty and differentiation in rationales for achieving precision and accuracy on the regulated matter and the conditions of protecting a market function such as competition.

It has to be taken into account that the law-and-economics (L&E) followers examine what legal rule will best promote economic efficiency and, on the other hand, originalists in constitutional interpretation and textualists in statutory interpretation (O&T) followers discern the original public meaning of the relevant legal text.¹⁵³ Hence, it could be said that

O&T is legal formalism and L&E is economic formalism, denoting two formalisms, and a judge stereotypically will reach the same result using either L&E or O&T, since L&E and O&T operate as mechanisms for justifying results reached on other, unstated and normative, arguments. Anyway, the law's content is adequately determinate to stipulate guidance in permitting the law to play its central coordination purpose.¹⁵⁴ Moreover, Congress legislates clear interpretive rules.¹⁵⁵ In *Pacific Bell v. linkLine*¹⁵⁶ the US Supreme Court has followed economic formalism by rejecting price squeeze liability, and so, the linkLine decision rests on the interaction of two legal doctrines: the absence of a duty to deal and the right to price at any level above marginal cost.

Are O&T and L&E neutral, objective, and determinate? Do O&T typically direct judges to apply L&E? What counts as a cost that must be encompassed in the analysis? Neil H. Buchanan and Michael C. Dorf¹⁵⁷ argue that "L&E and O&T typically obscure, rather than substitute for, normative value judgments." Statutory language like "appropriate and necessary" does not rule out cost-benefit analysis, but it hardly demands such analysis.¹⁵⁸ To that extent, L&E itself, involving cost-benefit analysis, is often unspecified. Todd Zywicki¹⁵⁹ believes that L&E and O&T are not just compatible but complementary. Additionally, L&E and O&T are both formalistic in the substance that they intend to generate results without reference to normative considerations, and so, formalism is exploited in order to have the best outcome in resolving disputes on a particular matter. In this context, Maggie Gardner¹⁶⁰ debates for certainty on the reason that where there is ambiguity, uncertainty, and subjective stimulation, judicial selection is complicated by subjective judging.

On the other hand, the economic theory underlying L&E is intrinsically subjective, utilizing a policy or set of market interactions grounded on a set of assumptions that is no more objective than any other set of assumptions. Hence, there is a need for utilizing the proper legal baselines that govern and enable market interactions. To that extent, Neil H. Buchanan and Michael C. Dorf¹⁶¹ argue that

"L&E and O&T provide objective-looking approaches that can in fact be adapted to any subjective priors that a scholar or

judge might have, it would be possible for liberals and progressives to purport to have discovered objective theories that quite reliably produce left-leaning results.”

It is characteristic that the core antitrust statutes are very brief and imprecise¹⁶² and the Supreme Court¹⁶³ rationalized that “[t]he legislative history makes it perfectly clear that [Congress] expected the courts to give shape to the statute’s broad mandate by drawing on common law tradition.” Moreover, the antitrust statutes are broad and general, and antitrust law applies to almost all viewpoints of commerce, but Judicial decision-making in antitrust has to adjust to any new technologies, business models, and market circumstances. Hence, existing principles can be adjusted to new business practices and market circumstances. To that extent, Justice Scalia¹⁶⁴ commented that “[N]o statute, can be entirely precise, and...some judgments, even some judgments involving policy considerations, must be left to the officers executing the law and to the judges applying it.”

Can formalistic rules undermine the objective of antitrust laws? It could be said that formalistic rules promote the objective of antitrust laws, which is concentrated on advancing a competition process based on certain ex ante legal rules that generate certainty and predictability which means that the rightful application of competition law will result in good economic outcomes for all parties in the market. By setting consumer welfare as the goal of antitrust law, which is a vague term encompassing an endless scale of values, it opens the road for various methods of achievement regardless of any certain legal rules denoting the anarchic way of arriving at economic results regardless of an equal competition environment for all parties in a market, leading to economic monopolies and market power.

It is characteristic that as economic analysis gradually informed antitrust law and policy, both courts and enforcement agencies recognize, on economic grounds, that some set of agreements must remain beyond Section 1 scrutiny. Hence, although the formalistic model of Section 1 embraced by *United States v. Yellow Cab*¹⁶⁵ entailed courts to make such examinations, it is gradually acclaimed that the condemnation of intra-enterprise conspiracies is deeply “orthogonal” to the fundamental antitrust objective.¹⁶⁶

It has to be taken into account that the economic analysis is displacing previously long-standing, formalistic models in antitrust, ranging from merger analysis to the use of vertical restraints and vertical integration. It is argued that the Sherman Act made a fundamental distinction between “unilateral” and “concerted” conduct and refused to condemn “coordinated conduct among officers or employees of the same company” which means that parent and subsidiary corporations set up a “single economic unit” that possessed “ultimate interests...[that] are identical,” the officers of each corporation were not “separate economic actors,” rendering Section 1 inapplicable.¹⁶⁷ On the other hand, a single corporation contains many competing interests leading to the collapse of many companies’ running.¹⁶⁸ It is worth mentioning here that while the court heavily presumes that intrafirm decisions are independent and driven by the boosting of profits, “in rare cases, that presumption does not hold”—as it did not for NFLP.¹⁶⁹ Hence, NFLP simply functioned as an instrument for the separate team decision-making, rather than acting as a single source of economic power itself. Justice Stevens¹⁷⁰ maintained that the court “eschewed such formalistic distinctions” as to whether the parties are “legally distinct entities” in deciding whether behavior establishes “concerted action” and so illustrating a rule for considering an integrated multi-component company as a single actor for reasons of Section 1 and removing the unity of interest inquiry away from incentive alignment and toward control rights, in line with the economic theory of the company.

On the one hand, if the NFL is just a “single entity,” its teams are immune from liability under § 1 of the Sherman Antitrust Act, which regulates combinations, contracts, and conspiracies in restraint of trade. On the other hand, if not, its teams possibly face antitrust litigation over the reasonableness of their cooperative activities due to the fact that a central distinction in antitrust law between “concerted” and “unilateral” conduct is the underpinning for the “single-entity doctrine” of dividing defendant businesses—all of which comprise multiple persons—into “single entities” immune from § 1 liability and cooperative arrangements among multiple entities subject to § 1 scrutiny.¹⁷¹ In *Monsanto*, it was decided that unilateral conduct by a single company acting independently is “not proscribed” by § 1.¹⁷²

Antitrust law makes a fundamental distinction between “concerted” conduct among multiple entities and “unilateral” conduct of single entities. Under the “single-entity doctrine,” § 1 of the Sherman Antitrust Act is inapplicable to the latter. Moreover, it is argued that the problem stems from the unavoidable fact that the “agreement” requirement of § 1 is itself a formal rule, not an economically meaningful standard. To that extent, the tension between such a formalistic requirement and an increasingly economics-based antitrust jurisprudence has resulted in the Supreme Court not expressed a clear and predictable standard for the application of the single-entity doctrine over the last century. To that extent, J. Schmitten¹⁷³ argues that

“a solution that embraces the formality of § 1’s “agreement” requirement with a formalistic rule for determining business organizations’ single-entity status. Under this approach, whether a business constituted a single entity for § 1 purposes would depend entirely on the “form” into which that business was organized. The courts would simply choose the “forms” into which persons must formally organize themselves in order to obtain § 1 immunity and then use antitrust regulation of mergers to police self-sorting into those forms. This would reduce the indeterminacy of current doctrine, which should appreciably reduce the administrative costs of antitrust litigation without a comparable risk of an increase in error costs.”

Hence, concerning the formalism of § 1’s “agreement,” the Supreme Court has to develop an exclusive list of corporate forms that constitute “single entities”¹⁷⁴ becoming a formalistic solution to the current doctrine’s indeterminacy.

It has to be taken into account that strictly general formalistic reasoning without any deep formalistic interpretation of the entire scale of the formal rule in antitrust often generates perverse results¹⁷⁵ and in line, J. Schmitten¹⁷⁶ argues that “Formalism—used here to refer to an antitrust

policy explicitly concerned with ‘form’ rather than ‘substance’—is a dirty word in antitrust scholarship.”

Antitrust formalism consists of commitments to interpretations of the antitrust laws that oblige courts to discount and even ignore relevant competitive effects. Thus, it involves the use of rigid rules resting on premises that are correct under some settings but not all, denoting the demand for a formalism that encompasses an effort to include all possible understandings of the matter involved in the formalistic interpretation of the formal law. Examples of antitrust formalism include per se rules, the analysis of collusion, the interpretation of the distinction between horizontal and vertical restraints, the “direct-purchaser” doctrine, and the per se rule against price fixing.

It could be said that competition law rules that downplay competitive effects appear to run afoul of the objectives of antitrust and, as such, antitrust formalism is counterintuitive because antitrust formalism involves commitments to interpretations of the antitrust laws demanding courts to discount and even ignore relevant competitive effects.¹⁷⁷ To that extent, Barak Orbach¹⁷⁸ argues that “in antitrust law competitive effects should matter and their exclusion through formalism is unsound.” However, implementing formal rules in a formalistic way that encompasses all the settings fitting into the particular formal term will have the required outcome settled as the goal of the legislature in passing the particular law, such as in competition law, which will be the result of the formal implementation of the rule and not looking for favorable economic outcomes in order to argue that there is rightful competition enforcement. This is because the economic outcome has to come under the implementation of formal rules favoring equally the whole market and not favoring the parties that have achieved better economic results regardless of the legal environment under which those economic results have been achieved. In line with this, K. Khoo and A. Sng¹⁷⁹ argue that “the ‘formalistic’ approach stands in contrast to its “effects-based” counterpart, where the authorities are required to establish actual anticompetitive effects in the relevant market.”

In a “formalistic” approach to Competition Law, liability is often ascertained pursuant to presumptions of law permitting the authorities to

infer liability upon proof of certain conduct; thus, the “formalistic” approach stands in contrast to its “effects-based” counterpart, where the authorities are necessitated to ascertain actual anticompetitive consequences in the relevant market. A formalistic approach should not only favor rigid rules but also more flexible and context-specific standards evaluating competing interests in a formalistic way, thus allowing the best solution for the competition process in achieving the best outcome for the market. It must be taken into account that adherence to the rule of law means that law enforcement is predictable, bringing legal certainty¹⁸⁰ which means that competition law is predictable.¹⁸¹ M. Schwarzschild¹⁸² argues that

“Predictability ranks fairly high among the legal virtues. It is part of what people mean by the Rule of Law. When the law is uncertain, human autonomy itself suffers, since without knowing what is permitted and what is forbidden it is less safe to act at all”.

On the other hand, Marcin Matczak¹⁸³ considers that judicial decisions based on formalistic reasoning challenge predictability instead of promoting it. Moreover, competition rules are adopted in order to prevent certain harms resulting from a restriction of competition, and this is achieved via ex ante principles of competition law discouraging and preventing companies from engaging in harmful conduct by making it illegal.¹⁸⁴

Formal per se rules generate legal certainty, but the interpretation of the content of formal rules has to be extensive in order to encompass all possible rationales expressing actual contacts in the field as part of the interpreted formal content of the formal law, denoting that formalism does not mean exclusion of any new rationales fitting the content and context of a formal law without counting on vague economic effects. Principles count first, and the outcomes of their implementation will fulfill their goals which in the case of competition law is generating an environment of fair competition grounded on a defined competition process based on rationales and doctrines. This means legal certainty and predictability, which is the main scope of the law as well. To that extent, Frederick Schauer¹⁸⁵ argues that

“I do not argue that formalism is always good or that legal systems ought often or even ever be formalistic. Nevertheless, I do want to urge a rethinking of the contemporary aversion to formalism. For even if what can be said for formalism is not in the end persuasive, the issues should be before us for inspection, rather than blocked by a discourse of epithets.”

On the other hand, if there is no decision according to the rules, then what is the role of having rules if their content is not applied as it is expressed by the legislature? Schauer says that the charge of formalism implies “narrowness,” but there is a need for “narrowness” because precision brings forward particular wording, and in any other case, vague wording and incorporating a range of events and notions leave the disputed matters unresolved, leading to a divergence, uncertainty, and unpredictability of what is rightful.

It is argued that formalism in *Lochner*¹⁸⁶ is focused on the absence of reference to political, moral, social, and economic choices implicating a direct reference to the content of the applied rules; thus, the judge is explaining a clear-cut statutory scheme. Of course, the interpretation of the content of a rule according to standard features denotes formalism under the understanding of applying formal rules and not looking for sporadic elements scattered from various occasions in order to give an alternative interpretation, which will not only fail to solve the disputed matters but also create a vague understanding, leading to uncertainty about what is rightful. In other words, even if there is room for debate about the application of a general term to a particular case, the acceptance of a very broad interpretation might lead to a situation where there is difficulty in finding the precise content and context for applying a rule, thus generating uncertainty and unpredictability for a particular matter such as in competition cases and so that the parties will not know what constitutes uncompetitive conduct, which means damaging the enforcement of competition rules and hinders the achievement of a fair competition environment.

H.L.A. Hart¹⁸⁷ certifies that legal terms have a substance of settled meaning and an unsettled meaning, and so Hart regards formalism as the

refusal to acknowledge, in cases of doubtful application, that the wording goes far beyond simply ascertaining the meaning of a word. On the other hand, on many occasions there is a need to formalize the content of the wording of a rule in a precise manner in order to achieve legal certainty and predictability of the law, rather than relying on effects of the employment of a rule, which might be too numerous, leading to many divergent judgments of a disputed matter. In other words, formalism is the choice of endorsing and applying certain legal rules defining a matter such as fair competition or anticompetitive contracts, and not encompassing various outcomes from different but correlated subjects in order to have a judgment accepted by particular interests to satisfy any political or social wishes against the rightful application of the endorsed rules. It is worth mentioning here that the principle of drafting is to translate policy into legislation; drafting is a technical skill that is reducible to a body of rules, which plainly denotes that politics, via governmental party policies, are forming the content of any law which means that politics affect the formation of any law not to mention the impact of politics upon the enforcement of laws.¹⁸⁸

Schauer says that the application of “the norm involves a choice not determined by the words of the norm alone,” but there is a demand for defining the norm and not allowing an overspread of meaning to be enclosed in a norm regulating a particular matter and so requiring a formality achieved via the standardization of the wording of the norm and its context, which means that in any other case there will be an endless discussion of the breadth of the norm, continuously accumulating elements in order to interpret the norm.¹⁸⁹ It is characteristic of the utilization of economic terms and outcomes in the case of competition, where too many economic theories can be exploited in defining or outlining fair competition, pro-competitive, and anticompetitive contact. In other words, ex ante formal rules defining all matters of competition and the competition rules are required in order to achieve rightful competition enforcement and not to employ economic or social principles to achieve political outcomes concerning competition by baptizing them as competition rules instead of endorsing a legal formalism of standardized legal rules.

Does formalism mean limitation of choice? On many occasions the court’s decision abides by the statutory language and, therefore, restricts the

choices open to the judge, which means following a formalistic application of the law in order to achieve the precise demands of the legislature on a particular dispute. Can a formalistic decision-making process be condemned as long as it complies with the formal law? Of course, resolving a disputed matter by following the clear wording of a formal law does not mean an unfair judgment¹⁹⁰ and, on the other hand, encompassing many other principles grounded in economic or social principles in the interpretation of the content of a rule claiming an adjustment in the context of the rule often leads many times in judgment not achieving the goal of the legislature but merely going along with politics rather than establishing a legally rightful precedent. It is essential to be “formalistic” and, therefore, endorse a narrow reading of a term in order to achieve certainty in regulation and real application of the law rather than contributing to a lax application of the law, which means not achieving the real goal of the rule. In the case of competition, the goal is that of implementing competition rules in defining the essence of competition, the competition process and anticompetitive conduct, and not merely to accumulate economic elements in order to satisfy consumers as a political task, which could lead to monopolies rather than implementing fair competition in a market open to all candidates.

It has to be taken into account that judgments are mechanically deducible from the language of legal rules because the law is a “closed system,” not allowing the encompassing of any factor as part of its content under the mask of interpretation which means that a judge cannot be condemned for a formalistic approach in merely applying the rightful perception of the employed term and thus accomplishing legal certainty and predictability as it has to take place in competition cases. It is characteristic that Judge Posner¹⁹¹ assumes that formalism is inapplicable to statutory rules for the reason that the decision to take these rules plainly is itself a choice in order to achieve the goal of the law and not endorse vague interpretation, in the name of not being formalistic, leading to divergence of standpoints and inapplicability of the real meaning of the law. Thus, it could be said that “formalism” entails “deductive logical reasoning” in applying the law and deriving from the content of the formal rule and not employing factors from various fields in order to relax the content of a rule, as it takes

place in competition by employing economic theories in implementing competition law. Finally, Judge Posner concludes that formalist reasoning has no application to the interpretation of canonical texts.

It is argued that legal formalism¹⁹² is better than other modes of construction for the reason that it best assists the reliance, expectation, enforcement, and other concerns characteristic of the given legal relationship. For instance, strict rules of construction in contract interpretation on functional grounds are promoted.¹⁹³

It has to be taken into account that legal systems inevitably necessitate the interpretation of regulatory language¹⁹⁴ regarding the rationale of the regulation, which in competition law has to be fair by defining the competition process and anticompetitive conduct rather than endorsing economic profits for consumers. This is necessary in order to define fair competition via the implication of economic factors and theories that do not lead to convergence and legal certainty, rather than a lax understanding of competition permitting monopolies to flourish.

Law is a mechanism for accomplishing social purposes, but in some situations, there is a need for general abstract rules in order to precisely define the regulated matters, which means that it necessitates a formalistic approach in resolving disputes on the particular matter. Formalism is an approach tolerable to the undertaking of correctly resolving problems confronted in law. Moreover, “Formalism” involves legal positivism, meaning that law is well-defined under formal rules, distinct from “morality.”¹⁹⁵ In addition, Paul N. Cox¹⁹⁶ says that “Law...is an instrument of social policy to be used for socially desirable ends,” but on the other hand, the content of the law has to be specific in order to regulate a particular matter and not illustrate a vague social term demanding vague moral explanations like sociology. Hence, law is specific to situation types or categories, incorporating the norms of real people in the real world¹⁹⁷ which are formalized and standardized to the possible degree in order to achieve legal certainty because the endorsement of a sociological analysis or economic one for legal matters is not leading to legal predictability. Thus, it has to be taken into account that legal interpretive rules must be durable which means that their application be predictable, neutral, consistent, and objective. In fact, legal formalism encloses predictability,

stability, and certainty. Hence, formalism is keeping the rule of law, stability, and certainty in legal judging.¹⁹⁸

It has to be taken into consideration that law prevalently directs activity and, therefore, law is dynamic in a dynamic society, but always in a definite context in order to achieve legal certainty and predictability, thus touching its formalist conception. Of course, social or technological conditions are changing, demanding new and different laws; however, its content has to be formally precise, enclosing legal principles that are formally defined and not being social terms that demand sociological or economic analysis. Even the content of legal rules, such as in competition law, can encompass economic terms that are specifically defined and do not allow imprecise interpretations, which means different judgments and thus a non-converging outcome leading to legal uncertainty.

It could be said that formalism denotes that the law consists of rules appealing to rule of law¹⁹⁹ values which means that rules allow the regulated subjects to facilitate the prediction of the legal effect. Hence, formalist adjudication means not departing/deviating from the known meaning of a rule and so not undermining or destroying the reasons for rules, such as in the case of competition, where the prevalence of economic analysis in deciding the implementation of competition law has led to a lack of strong competition enforcement via a divergence in defining anticompetitive conduct, fair competition or determining the competition process itself, allowing the emergence of monopolies instead of an open market for all prospective candidates to engage freely in trading.

Does formalism characterize an uncritical reliance upon the “facts” found in legal proceedings? Formalism entails fact-finding, and objective rules are favored over vague standards, as standards necessitate or enable the assessment of more facts and thus give primacy to formal rules via factual evidence. Hence, formalists not only prefer abstract and general rules but also particularized or specialized rules, as long as their content can be defined precisely, not allowing a vague interpretation leading to multiple judgments. Real formalism does not encompass the suppression of facts; on the contrary, formalism gives facts the meaning echoed from formal rules and not vague social norms. Additionally, formalism does not implicate vague economic theories, and subsequently, definite economic factors are

taken into consideration in an economic formalism approach, which can be employed by legal formalism which means that formal economic rationales can be incorporated into legal rules implemented in a formalistic way. Paul N. Cox²⁰⁰ says that “formalism is both viable and, at least to me and perhaps a few others, attractive...Formalism requires restraint in the form of a tolerance of apparent injustice, apparent absurdity, even apparent evil.” On the other hand, formalism means implementing the law in a way that brings forward standardized standpoints dealing with the implementation of the tasks of the legislature in strictly legal terms via legal rationales and not applying social understandings in a vague way in order to fulfill political and economic ambitions, thus arriving at divergent judgments, which means not achieving legal certainty and predictability. Hence, formalism does not denote unfair justice or judgments grounded on unsound rules.

A useful example of utilizing formalism is the investigation of negotiable instruments, and the formalist position asserts that the question of whether any person in possession of a negotiable instrument is a holder in due course necessitates comparatively clear and formal demarcation standards, for the reason that holder in due course status is a weighty consideration for anyone who thinks of taking an instrument as a cash substitute or a credit instrument. To that extent, Merrill and Smith²⁰¹ comment that “negotiability imposes very strict formality requirements precisely in order to reduce the need to measure the reliability of an instrument.” In parallel, it could be said that in the case of competition, the competition law rules have to be clear, requiring relatively distinct and formal demarcation standards concerning the factors taken into account in order to define anticompetitive conduct, defining competition process and fair competition so as to achieve legal certainty in reaching rightful competition enforcement. In other words, there is a need for a formalist architecture of rules. Furthermore, there is a need to understand formalism in a manner that, while retaining the importance of form, does not disconnect it entirely from the functional context of relations.

Legal formalism and linguistic formalism are struggling for systematic perfection of expression or performance, or both, and so, formalism maintains the ex ante value of systematic and general, clear-cut, bright-line rules stimulating predictability and efficiency. Also, the justification of

formalism that underlies Broadway Management is that in an area of practice assessed by a demand offering certainty.²⁰² It could be said that a stricter formalism is required in some areas of private law, such as in competition law, rather than other areas. In line with this, J. Yovel²⁰³ argues that

“Relational formalism emerges as formalism in practice, where determining what belongs to the category of form and what belongs to context is itself a matter of application and construction...The court that values adherence to valid, binding legal rules and understands its own role in terms of enforcing them is the court that applies strict formalism... Relational formalism emerges from this study as a practical operative jurisprudence that is preferable both to strict formalism and to all-out realist adjudication.”

It could be said that formalism is a useful interpretative approach for competition law. Formalism is a device serving the interests expressed by the law and not superseding those interests, which in the case of competition law is translated into promoting a standardized anticompetitive conduct and a defined competition process for rightful enforcement of competition rules in order to achieve fair competition and not monopolization of the market.

It seems that formalism is grounded in the proposition of the “rule of law,” which takes for granted the law’s capacity to manifest the diverging interests in society. On the other hand, it seems that the ruling elite has relied upon the “rule of law” to authorize its power and resist “would-be usurpers”²⁰⁴ but the rule of law should promote the real interests of a society and not merely the interests of an elite against the good of people/society. Furthermore, formalism is grounded on the fact that law can be understood from within its formal content, which means that formalism necessitates an interpretive method that relies on the text of the relevant law while lessening extratextual sources of law, thus generating legal certainty.²⁰⁵

It has to be taken into consideration that law produces social order, which means that law must be practically effective, and so, law is capable of generating and sustaining order. Moreover, the rule of law denotes deliberative formalism due to the fact that law functions as an authoritative source of normative guidance, and so, the law must be fit for use as such, which means that in order to be guided by law as an authoritative source of normative guidance, an engagement with it in a deliberatively formalistic way is required essentially. Furthermore, law is supposed to be an authoritative source of normative guidance offering effective normative assistance by concentrating on the structural formality of law along with the deliberative formality of legal reasoning. Additionally, the new formalism²⁰⁶ regards the formality of law as an implication of legality and the rule of law.²⁰⁷ Moreover, Paul B. Miller²⁰⁸ argues that

“The structural formality of law and deliberatively formalistic legal reasoning are thus the mechanisms through which the law generates and maintains order, even as the relationships between the content of law, responsive deliberative engagement with law, and behavioral manifestations of order are neither fixed nor linear.”

The unpredictability of vague standards stimulates actors to look for clearer rules offering predictability in the litigation process. It is argued that legal formalism takes many forms, and *Lochner*²⁰⁹ is referred to as a classic example of formalism. Hence, the decision is based upon formal rules defining the disputed matter, and the court’s reasoning was formalistic by considering the term “liberty” as self-defining.

It is argued that “formalism” designates the mechanistic application of rules without consideration of their purposes, and so, it could be said that formalism focuses on the separation of law and politics.²¹⁰ H. Kislowicz and B. Berger²¹¹ argue that

“at a general level, formalism is designed to remove at least some measure of discretion from decision-makers...”

Formalism can inhibit courts from making some of their reasons for decision explicit, damaging the transparency that is essential to structures of legitimacy.”

On the other hand, formalism allows the judgment to be grounded in the content of the law, which regulates entirely the disputed matter, thus permitting the elaboration of the legal reasoning.

It is argued that formalism prevents courts from remedying injustices that do not fit the pre-defined categories; on the other hand, courts cannot become regulators by adding new content to a law and so, it is not a matter of formalism but a matter of amending the law by the legislature. Formalism does not mean that we cannot gather evidence in interpreting the formal content of the law, but it means that the judgment is grounded in the formal content of the implied rule and not any other fact outside of the precise content of the rule. Hence, it cannot be said that formalism is getting in the way of principled justice.

It is characteristic that an increase in the citation to nonlegal information in judgments signals a break in acceptable authority, indicating the declined dominance of the traditional role of legal information and, in turn, generating the phenomenon of “delegalization” of law.²¹² Does resolving matters of competition law via economic effects analysis lead to “delegalization” of law? Besides, the judgment has to be grounded in the law and not on nonlegal information, which merely can be utilized in interpreting the content of the law. Moreover, formalism stresses the rational deducibility of law.²¹³ Also, Richard Posner²¹⁴ portrayed legal formalism as the theory that “once lawmakers produce rules, judges should apply them to the facts of a case without regard to social interests and public policy.” In addition, David Lyons argues that formalism is a manifestation of the competence of law as a means for deciding any case that occurs, and the law is “complete and univocal.”²¹⁵ It could be said that formalism stresses the role of the judge and the adjudication process as the autonomous way of deciding the one and only outcome of a case, undiluted by nonlegal normative aspects of law or political thinking. In contrast, realism underlined the sociological reliability of a decision.²¹⁶

It has to be taken into account that formalism does not mean applying a “rigidly doctrinaire”²¹⁷ type of syllogistic reasoning, causing a misapplication of law principles to generate socially unacceptable results²¹⁸ and consequently injustice. Hence, formalism denotes a mindful rule application characterized by implementing normative principles internal to law.²¹⁹ It could be said that “doctrinalism” is honored by formalism, involving the core of a judgment and the characteristics of adjudication.²²⁰ In addition, formalism is impartial for the reason that it utilizes legal doctrines, thereby embracing canons and principles choices.²²¹

Taking into consideration that lawmakers make the law predictable, Justice Gorsuch²²² specified that it is not legitimate “the judge ... to revise the law willy-nilly” which means that judges should refine a law’s pre-standing sense. In other words, rules should plainly produce the expected outcomes²²³ which means that laws have to be both predictable and restrictive. The rise of legal formalism is correlated with the accomplishment of substantive legal changes such as establishing the supremacy of the right of contract,²²⁴ and so formalism advanced the objective nature of the market by neutralizing and defusing the political and “redistributional” prospective of law, which means that legal formalism’s appearance motivated law toward embracing a standard of objectivity which means objective judgment and politically neutral application of legal rules.²²⁵ Likewise, in formalism, Pildes²²⁶ says that law is “an internally valid, autonomous, and self-justifying science” in which right solutions are “derived from the autonomous, logical working out of the system.” In line, Barak Orbach²²⁷ argues that “both rigid formalism that curtails the ability of courts to exercise discretion and informality that leaves courts with unguided discretion tend to lead to arbitrary outcomes and cannot persistently serve society.”

Formalism does not mean that the formal terms of law are interpreted in a totally strict way, but there is a need to encompass the entire spectrum of matters included in the understanding of the term, as long as the matters are defined under formal rules and are not being merely social stories, economic facts, or political ambitions. Therefore, if there are new developments due to technology, social, and economic advancements, the new understanding will be considered part of the content of formal law. It is

argued that rules and standards tend to represent degrees of discretion, not types of legal norms.²²⁸ On the other hand, legal norms are encompassed in formal rules, but the interpretation includes degrees of discretion in identifying the settings for the implementation of the norms in the specific environment and field. It is characteristic that formalism is utilized in defining the categories of per se liability.²²⁹

American antitrust law today is categorically rooted in economics. In this context, J. Schmitt²³⁰ argues that “the realities of statute-based law are such that antitrust law may always contain residual, economically unjustified legal formalities.” On the other hand, formal law cannot avoid formalities, but the exact definition of the factors defining the regulated matter adds to legal certainty and predictability, which means justified economic approach as long as competition rules are formalistic and not based on vague economic factors. This leads to chaotic and divergent approaches concerning the definition of the goal and the implementation of the task of competition law, which has to be strictly defined and not left under vague terms such as protecting the economy, but formal definition of a precise task such as facilitating an open market for all candidates to operate under equal opportunities and defining the exact legal terms of the competition process under which the goal is fulfilled.

By setting as the goal of competition law the growth of an economy or the good of consumers, it denotes the wish to have an unidentified goal in order to allow many views for promoting competition and countless economic facts contributing to divergent judgments. This enables capital to operate freely in a free way of achieving its expansion without any cost or order. It seems that the prevalence of an increasingly economics-based antitrust jurisprudence over statutorily imposed formalistic rules has led to a divergence in defining which actions are pro-competitive and which are anticompetitive. This has allowed the emergence of monopolies and oligopolies in many fields, not mentioning the inability to apply the same economic terminology and analysis to digital corporations and the functioning of the digital economy and the forthcoming AI economy has resulted in an unregulated digital economy untouched by competition and competition law and enforcement.

Taking into account that competition and the goal of competition law are supposed not to be altered since its appearance, it seems that from a formalistic rule in the *Yellow Cab*,²³¹ the court is moved to Copperweld's²³² anti-formalistic approach, denoting the legal uncertainty and predictability regarding competition law and enforcement.²³³

It has to be taken into account that the distinction between horizontal and vertical restraints is formalistic, energizing a dogmatic distinction between vertical and horizontal restraints, which means standardization of the concept and, thus, bringing legal certainty to competition. Also, antitrust formalism offers the structure for antitrust analysis and is significantly vital. Nonetheless, the device is often abused by forcing presumptions and rules to their extremes, and so, there is no need for a strictly rigid interpretation but one enclosing a precisely defined reality.

It is worth mentioning here that informality and formalism are not separate approaches, but noticeably they are two ends of a scale which means that some formalism is necessitated to structure a meaningful decision-making process and that exclusion of sources of information can harm the quality of the decision-making. Therefore, the actual selection is of the degree of formalism.

What is the relation of formalism and functionalism? Formalism is characterized by categorical rules, deductive reasoning, and an aspiration toward predictability²³⁴ and thus formalism is the court's preferred methodology in bankruptcy cases. In contrast, functionalism is marked by fact-intensive rules, inductive reasoning, and an aspiration toward efficiency²³⁵ and so, in preferring formalism, the court has underestimated functionalism.

Formalism values "transparency, predictability, and continuity," whereas functionalism respects "adaptability, efficacy, and justice in law." Moreover, formalism mainly supports "deduction from authoritative constitutional text," while functionalism upholds "induction from constitutional policy and practice."²³⁶ Formalists deem that final judgment and enforcement must come from Article III court.²³⁷ Chief Justice Roberts suggested that formalism is the only way to achieve the promise of Article III.²³⁸

It could be said that functionalism highlights the values of market freedom and competition, which are jeopardized by state intervention.

Hence, formalist law has to be functional with the purpose of surviving challenges arising from societal diversity, political controversy, and economic growth. Functionalism is an appropriate formula for law's capability to survive, primarily by remaining adjustable and responsive. In line with this, B. Srbinovski²³⁹ argues that

“functionalism has a stronger pedigree and is better suited to the scale and variability of the issues that come before modern courts in mass tort cases. A functional approach is not only better aligned with the substantive goals of bankruptcy, but also deeply embedded in the traditions of the practice of bankruptcy law.”

Furthermore, functionalism in law illustrates the manner in which the flexibilization and modernization of formal law, in response to a gradually multiplex social environment made up of competing interests and contestations, occurs if law is to preserve a steering role. In other words, good formal law is functional.

Do judges decide outside the law according to functionalism? Judges have to decide according to the wording of a rule, taking into account the perception of the disputed matter that prevails in commerce or the social sphere. Can courts ignore the formal law of the disputed matter and decide based on a practical approach? Courts have to apply the relevant formal law, otherwise any decision grounded in practice and not conforming to the formal law will be annulled.

Finally, it could be said that new technologies brought forth the demand for Neo-formalism, denoting a rationale that a principle has to be interpreted in such a way that substantial furtherance of the precise statutory or constitutional prescription must be endorsed in order to cover all the possible occasions fitting within the wording of the principle. Additionally, the AAI Neo-formalism will express the rationale of Neo-formalism and will be operated by AAI humanoids in an AAI economy where objectivity will be exploited in order for the whole AAI system to function properly, illustrating the new spirit of running applications because otherwise the same cons/pathologies of the conventional system will emerge.

It has to be taken into account that consciousness is the root of biological intelligence, but the development of AI reveals that intelligence can exist without consciousness. On the other hand, humans have entered a new era by making current robots to extend human intelligence, but the forthcoming AAI humanoids will be able to engage in all human biological and social activities, shaping the AAI society. Even current technology robots are anticipated to replace or even outstrip human intellectual work with competence and accuracy. Are humans expecting robots to have the same humans consciousness? Of course, AAI humanoids will have the consciousness suitable for allowing them to function properly in an AAI world and probably in both material and spiritual dimensions. Furthermore, when AAI humanoids have self-awareness, communicating directly like humans and developing collective values, then they will represent human “heterogeneity.”

Will humans generate a new type of life via AAI technology? Currently, AI is utilized by humans as a tool in advancing human capabilities in the whole sphere of life. With the forthcoming AAI technology and the appearance of AAI humanoids with capacities beyond humans, the AAI judge will interpret the law according to its wording, covering all possible occasions and taking into account of course general economic principles that elaborate on the general wording of competition law. Thus, according to the formalistic approach, AAI technology will categorize the occasions regulated by AAI competition law rules.

4.6 Finance and Competition

It has to be taken into account that the spread of FinTech brings both opportunities and challenges to the financial services industry. On the one hand, augmented FinTech activities stimulate greater competition and diversity in lending, payments, trading, and other financial services, leading to broader financial inclusion and a multiplied customer base resulting in enhanced revenue streams, upgraded efficiency, and enriched resilience in the banking system. On the other hand, heightened competition puts pressure on the profitability of conventional financial institutions (FIs)

taking on extra risks to maintain their margins. To that extent, Selim Elekdag²⁴⁰ et al. argue that

“higher capital and liquidity ratios as well as greater income diversification can reduce the influence of growth FinTech activities on risk taking across FIs. Likewise, there is suggestive evidence indicating that in some cases greater FinTech presence may be associated with less FI risk taking amid stronger institutions.”

The banking industry is experiencing a digital revolution, and so, by sensing pressure, FinTech corporations and BigTech platforms are endorsing AI technology in traditional banking businesses.²⁴¹ Hence, the digital revolution has had significant consequences for banking, and so, digital banking platforms, encompassing mobile applications and websites, have become prevalent. Thus, digital AI technology serves as an alternative to brick-and-mortar branch networks, becoming the leading means through which consumers access and banks provide financial services.²⁴² Thus, the digital revolution concerns how banks compete, and competition in banking is of particular interest as a result of its inferences on financial stability, influencing the whole economy. Digital banking alters both the size distribution of banks and the composition of services they deliver. To that extent, Naz Koont²⁴³ argues that “digital platforms have increased competition in deposit and loan markets by allowing mid-sized banks—those with relatively high-quality digital platforms but without extensive branch networks—to gain market share through higher demand and lower costs.”

Digital banks strengthen their share of uninsured deposits while downgrading their share of loan originations to low-income borrowers, somewhat driven by a lower capability to digitally screen or monitor this market segment. Thus, digitalization modifies the likelihood of bank distress by increasing funding instability as measured by banks’ share of uninsured deposits, and by leading to raised credit risks in market segments that are less well supplied by digital technologies. Additionally, digital banks face higher deposit outflows for each deposit market segment in

response to increases in interest rates, leading to a lower deposit value for a given level and composition of deposits.²⁴⁴ Hence, the increase in size and geographic reach of mid-sized banks makes them more significant from a financial stability point of view. Moreover, concentration, market integration, and markups augment competition in banking.

It is worth noting that the digitalization of banking has consequences for monetary policy because the spread of digital banking modifies banks' capability to screen or monitor customers across different market segments, elevating risks in market segments that are less well served via digital technologies. To that extent, Naz Koont²⁴⁵ argues that

“digital banks attract a different and potentially flightier deposit base. The 2023 banking crisis highlights the dangers of a deposit base that is mostly comprised of uninsured deposits...digital banking also reduces the variable costs of deposit provision... digital banks also face higher deposit outflows for each deposit market segment in response to increases in interest rates, leading to a lower deposit franchise value for a given level and composition of deposits.”

Increased competition among banks leads to heightened financial instability as intensified competition erodes market power, diminishes profit margins, and damages value. Thus, banks face greater enticements to engage in excessive risk-taking.²⁴⁶ It is argued that greater competition not only augments bank fragility but does so to a greater extent as activity constraints become more stringent. Hence, in banking markets characterized by amplified competition, banks typically maintain lower capital buffers, and the competition has a negative effect on banks' risk attitudes.²⁴⁷ Moreover, competition boosts financial stability by lowering loan rates, diminishing borrower credit risk, and reducing the possibility of bank failures.²⁴⁸ It must be taken into account that lower entry barriers boost lending and growth, but also risk-taking.²⁴⁹ Competition reduces the overall riskiness of bank lending, stimulating lower interest rates and, thus, moderating moral hazard concerns of bank borrowers.²⁵⁰ Due to the

existence of information barriers in global trade along with the digital revolution, Xuanyi Shi²⁵¹ et al. argue that “banks with foreign investors tend to create less liquidity, especially in on-balance sheet liquidity creation activities.”

It is characteristic that global trade has been affected by military conflicts, political pressures, and trade wars; thus, the global supply chain is facing restructuring tension. To that extent, Xuejing Yang²⁵² et al. argue that “In this political current, the US government is increasingly moving toward local protectionism. Another outcome of the trade war is the shift in the world’s personal value orientation brought about by the change in political winds.”

Financial institutions are not immune from antitrust laws; therefore, competition policy is part of federal bank regulation, overseen by the specialized regulatory agencies. US banking law ascertains competition policy in banking through regulatory review of bank mergers and acquisitions, anti-tying rules, and prohibitions on management interlocks. Thus, this mode signifies the direct, or formal, application of antitrust to banking institutions.²⁵³ The power of banks stems from the aggregate control of banks over credit.²⁵⁴

Upholding the applicability of the Sherman and Clayton Acts to commercial banking, the Court recognized that bank regulation sought to “minimize the hazards of intense competition.”²⁵⁵ Anti-tying in banking law restricts a practice viewed as anticompetitive in most business contexts and so, in the US, the Clayton Act outlaws tying arrangements that “substantially lessen competition or tend to create a monopoly in any line of commerce.” In the banking context, this policy apprehension is vital owing to banks’ legal monopoly on money creation.²⁵⁶

The banking sector is adopting information technology and Big Data in response to the increased supply of technology and to alterations in consumer expectations of service being drivers of digital disruption.²⁵⁷ Information technology allows financial intermediaries to advance personalized services. Additionally, risks are triggered by the privatization of big data, and it seems that privacy laws are unsatisfactorily protective of peoples’ data when private technology corporations collect it.²⁵⁸

Banks and fintech corporations channel almost all payments between agencies and the public via different programs and, consequently, progressively collect more data while doing so.²⁵⁹ To that extent, the digitalization process is promoting remote loan operations. Additionally, centralized digital infrastructures are vulnerable to software updates superseding original safeguards.

A bank's IT progress augments its competitive advantage. If IT progress involves only a drop in the costs of supervising an entrepreneur without shifting banks' relative cost advantage, then banks' competition force will not be affected by such technological progress. Moreover, when competition in the lending market is at a low point, intensifying competitive intensity enhances welfare because more competition boosts entrepreneurs' utility. On the one hand, expanding competition power assists entrepreneurs; on the other hand, more competition lessens banks' profits.²⁶⁰

4.7 Formalism in Financial Competition

It is characteristic that the Supreme Court applied antitrust rules by reference to effects on "competition" as such.²⁶¹ Hence, antitrust should turn away from the welfare and efficiency tests that have conquered antitrust law, returning to the protection of competition and the competitive process, which means that antitrust rules have to assess whether some behavior advances or hinders "competition."

Taking into account that the antitrust statutes are broad and generalized, courts have formed a system of doctrine in implementing the prohibition on restraint of trade in Section 1 of the Sherman Act, the prohibition on monopolization²⁶² in Section 2 of the Sherman Act, and the prohibition on certain mergers and acquisitions in Section 7 of the Clayton Act, which encompasses court-created rules relying on an unliquidated "competition" criterion.²⁶³ It is worth noting that Senator Sherman proposed the prohibition of agreements "which tend to prevent full and free competition," which has to be seen as the spirit of the legislature regarding competition.

Per se illegality applies to practices that are openly or clearly “anticompetitive,” and operative legal rules rest on whether some practice or transaction damages or promotes competition. Jonathan Kanter²⁶⁴ has suggested that courts embrace a merger test based on whether the “competitive process” has been damaged, rather than on any assessment of outcome consequences, which means that there is a need to concentrate on competition as an administrable standard rather than on quantifying consumer welfare results. The competitive process should be operationalized in legal doctrine in a formalistic way, leaving room for interpretation under other formal rules covering competition in order to achieve a standardized understanding of the actions boosting competition and, thus, defining the anticompetitive contacts. In line with this, Greg Werden²⁶⁵ claims that the Supreme Court’s rule of reason decisions “exhibit a single-minded focus on the competitive process,” and that “impact on competition” transpires as the “single criterion for evaluation [of] the legality of a trade restraint,” and so, there is reliance on an unliquidated harm to competition test.

It is argued that “competition” should be approached not via law but economics due to the fact that many antitrust concepts such as monopoly, substitutability, and so on are borrowed from economics, and so the evaluation is determinedly “welfarist.”²⁶⁶ Competition law regulates various matters such as fair competition, which have economic outcomes, but law doctrines govern the disputed matters, and so economic rules should not be considered as the legal rules governing these matters.

It is characteristic that the Court used “competition” broadly, describing market interactions free from antitrust violations and prohibiting actions such as abolishing the chance of buyers and sellers to contract on the basis of fair, open, free competition.²⁶⁷ Thus, courts often imply that whatever advances overall welfare is procompetitive, while whatever diminishes it is anticompetitive.²⁶⁸

It is worth mentioning here that the Supreme Court has denied that the competition criterion can be focused on welfare results, since a transaction can harm competition even if its effect on welfare is positive²⁶⁹ which means that feasible economies cannot be utilized as a cover for illegality. Moreover, the Court has underlined that welfare harm does not verify harm

to competition and so neither the charging of monopoly prices nor cutting off a rival was considered “anticompetitive.”²⁷⁰ From case analysis, it emerges that the “competition” criterion is grounded in a multipart and inconsistent relationship with a whole display of rather different values. Hence, antitrust should maximize and optimize fair competition in the market via legal doctrines encompassing standardized economic rationales. To that extent, Steven C. Salop²⁷¹ argues that “while antitrust law in principle can evolve, new legislation would be a more rapid—and more certain—path to reform.”

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